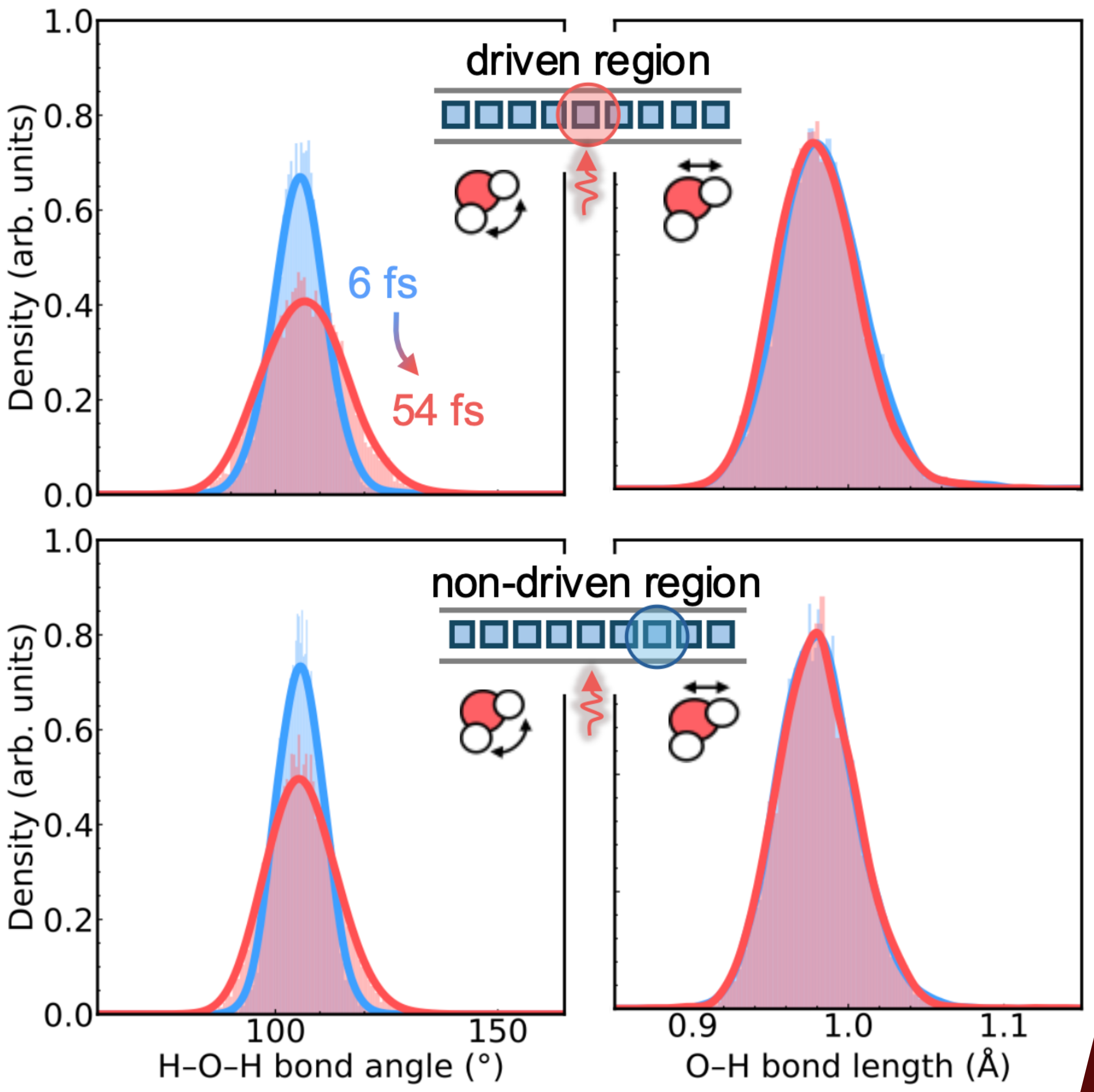
Light & Matter Research Group
at Texas A&M



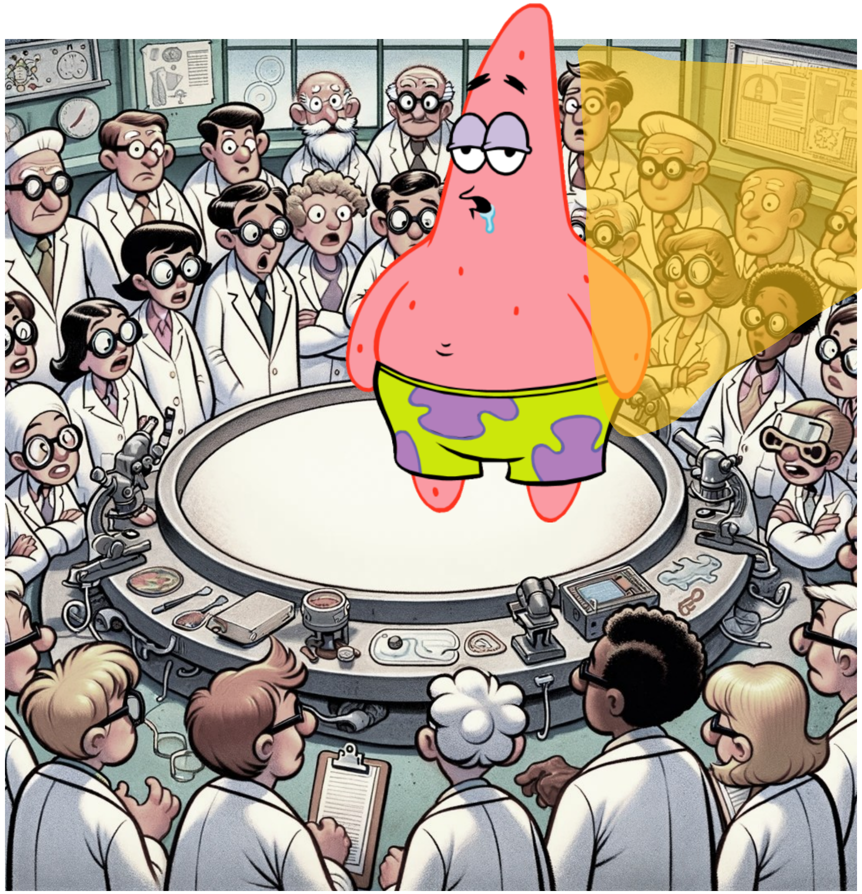
TEXAS A&M UNIVERSITY

Controlling Mode-Selective Energy Transport in Cavity-Coupled Water

Sachith Wickramasinghe

September 2nd 2026

Matter

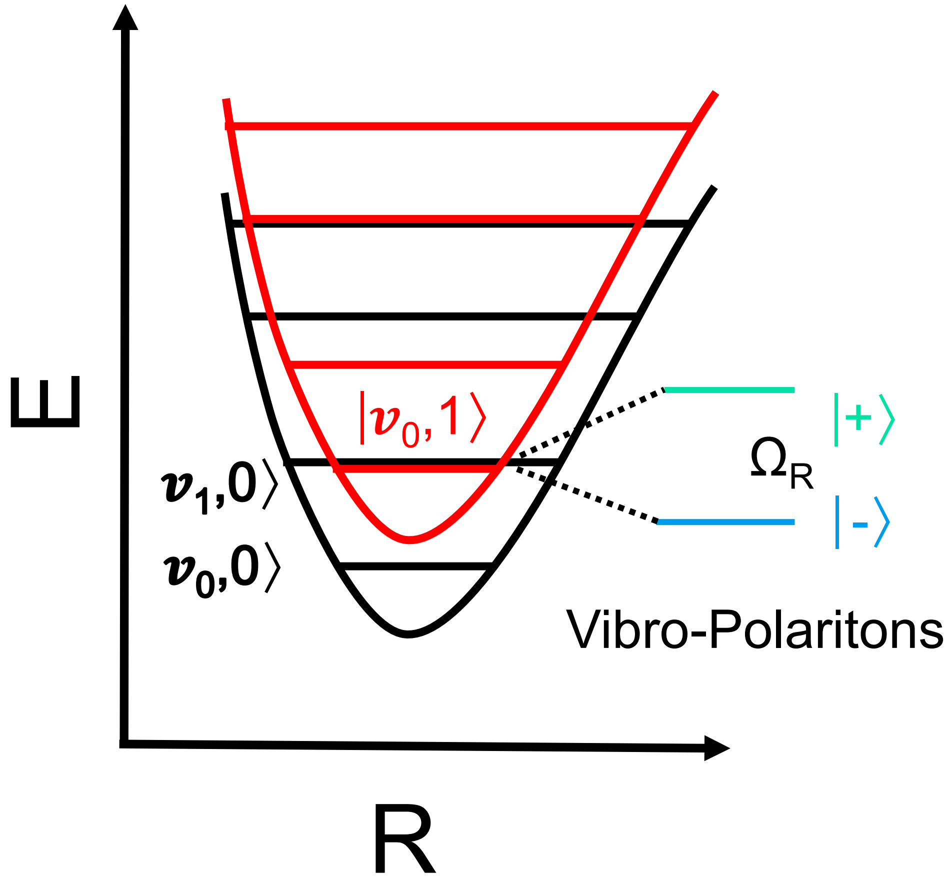


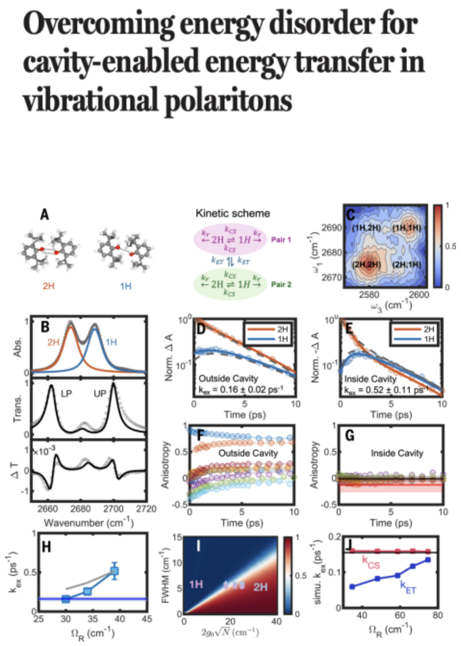
Matter

Light

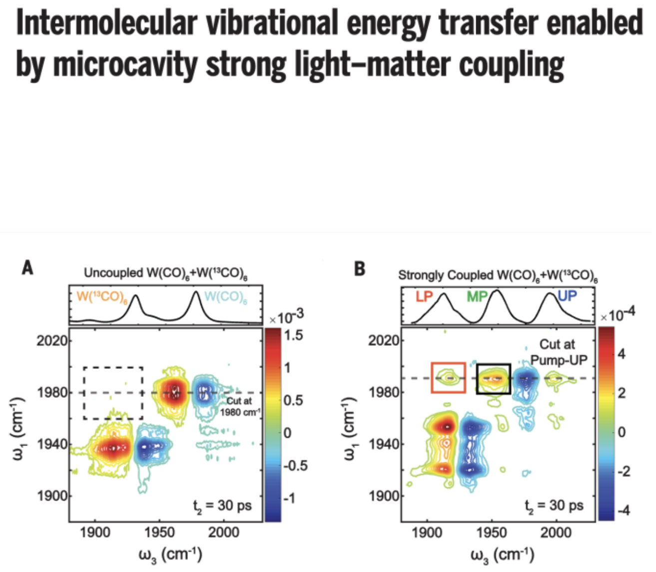
Experiments: Modifying Chemistry in the Dark

(Vibrational Polariton)





Yin, G., Liu, T., Zhang, L., Sheng, T., Mao, H., & Xiong, W. ***Science***, (2025).



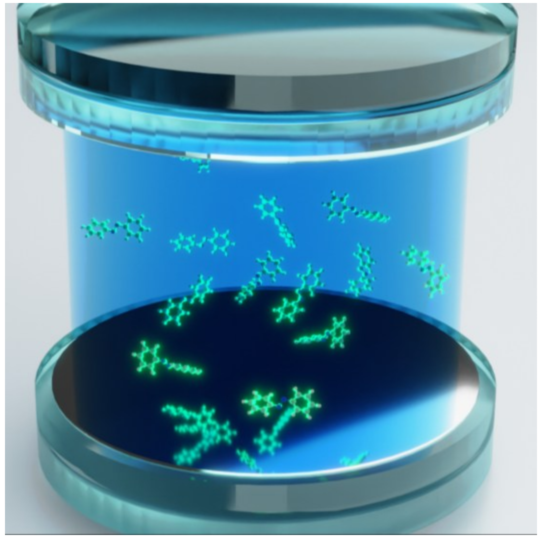
Xiang, B., Ribeiro, R.F., Du, M., Chen, L., Yang, Z., Wang, J., Yuen-Zhou, J. and Xiong, W. ***Science***, (2020)

Coupling vibrations to vacuum radiation modifies chemistry.

But how ?

Can Weak Molecular Coupling Modify Molecular Dynamics?

Each individual molecule is coupling very weakly via g



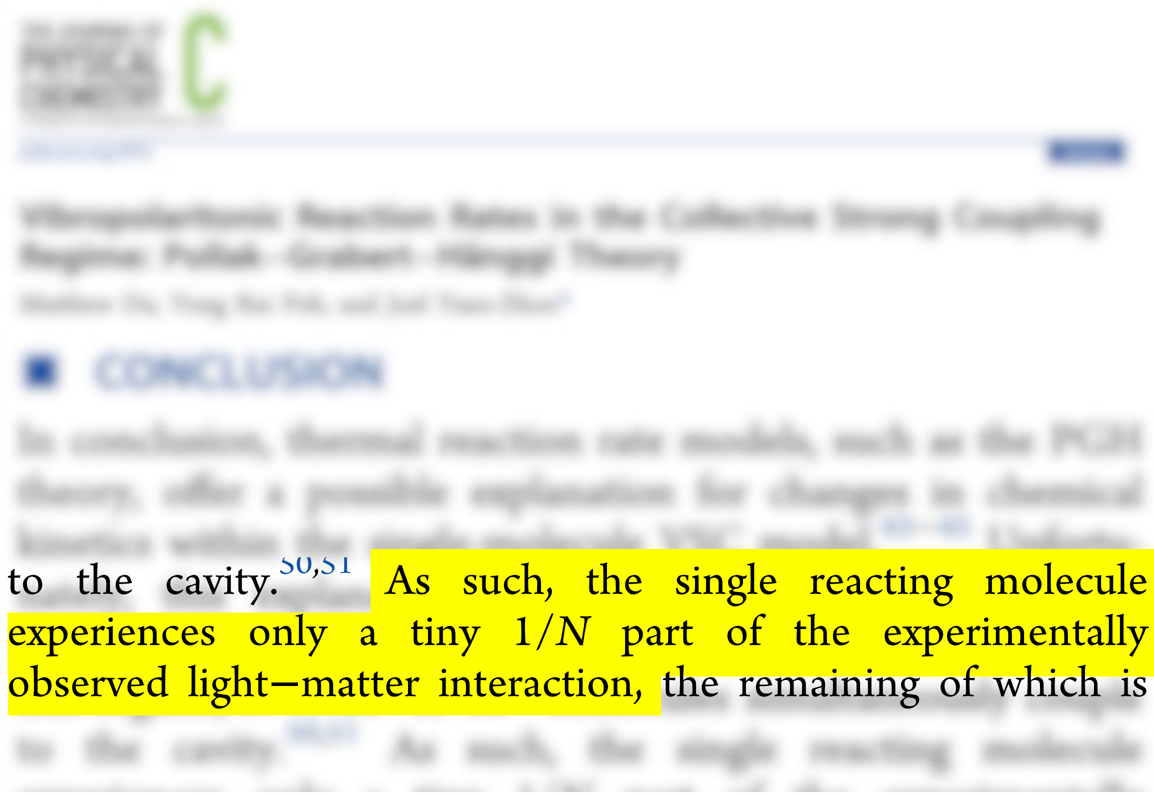
Rabi-Splitting = $\Omega = 2\sqrt{N}g$

Cavity couples to a macroscopic number of molecules via **collective coupling**.

Chemistry happens **locally.** From the perspective of a single molecule light-matter coupling is negligible.

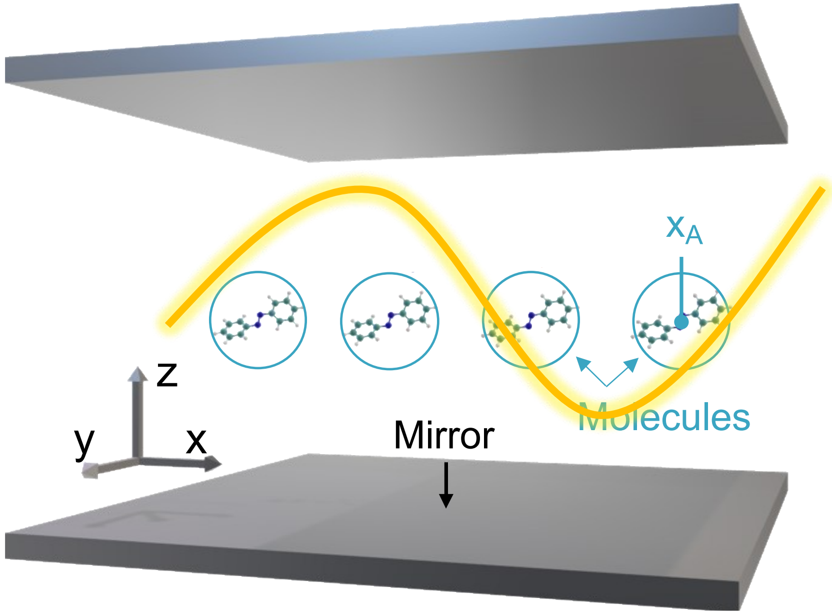
Most experiments couple a macroscopic number of molecules/atoms to quantized radiation.

$$N \rightarrow 10^{10} - 10^{23}$$



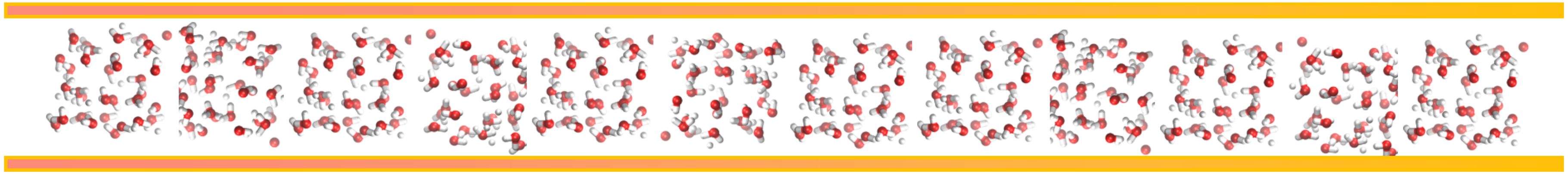
to the cavity.[50,51] As such, the single reacting molecule experiences only a tiny $1/N$ part of the experimentally observed light—matter interaction, the remaining of which is

What is needed?



- **Spatial variation of the cavity field**
  → beyond the **long-wavelength approximation**
- **Multiple cavity photon modes**
  → beyond the **single-mode approximation**
- **Intermolecular interactions**
  → essential for molecular energy redistribution
- **Mesoscale simulations are required**
  → to capture the collective spatial and dynamical effects

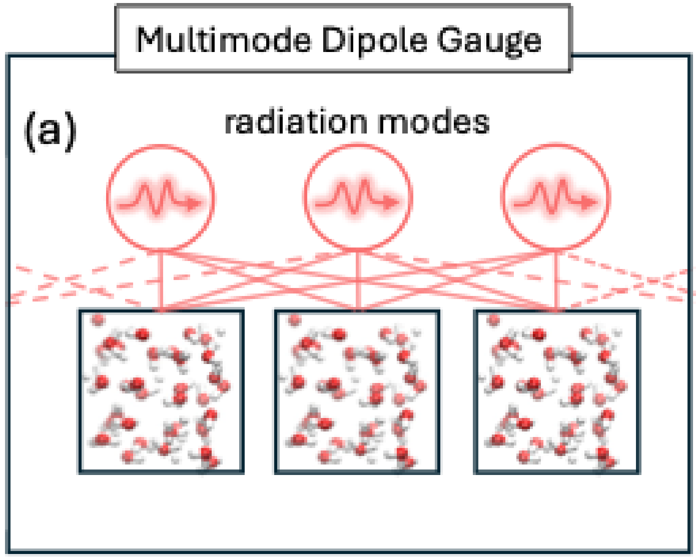
Theory



Multimode Dipole Gauge Hamiltonian

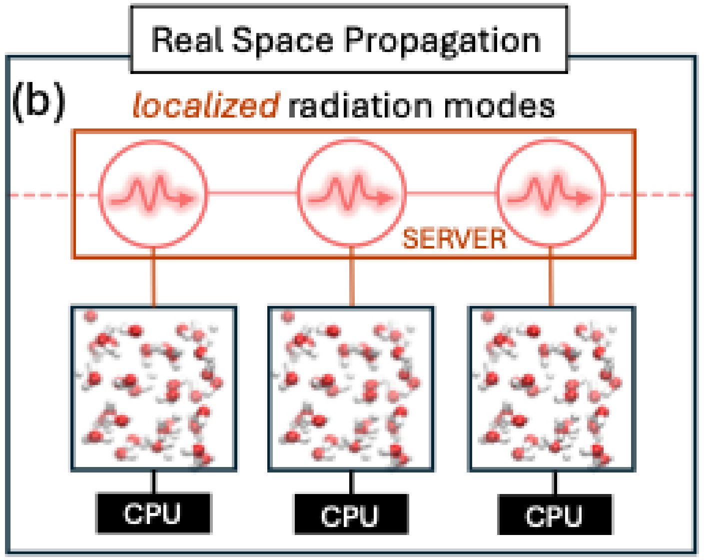
$$\hat{H}_{\mathrm{LM}} = \hat{H}_{\mathrm{M}} + \sum_{k} \omega_k \hat{a}_k^\dagger \hat{a}_k + g_0{}^2 \sum_{m,n,k} \mathrm{e}^{ik(R_n - R_m)} \hat{M}_m \hat{M}_n$$

$$+ g_0 \sum_{n,k} \sqrt{\omega_k} \left(\hat{a}_k^\dagger \mathrm{e}^{-ikR_n} + \mathrm{e}^{ikR_n} \hat{a}_k \right) \hat{M}_n .$$



Real Space Propagation Hamiltonian

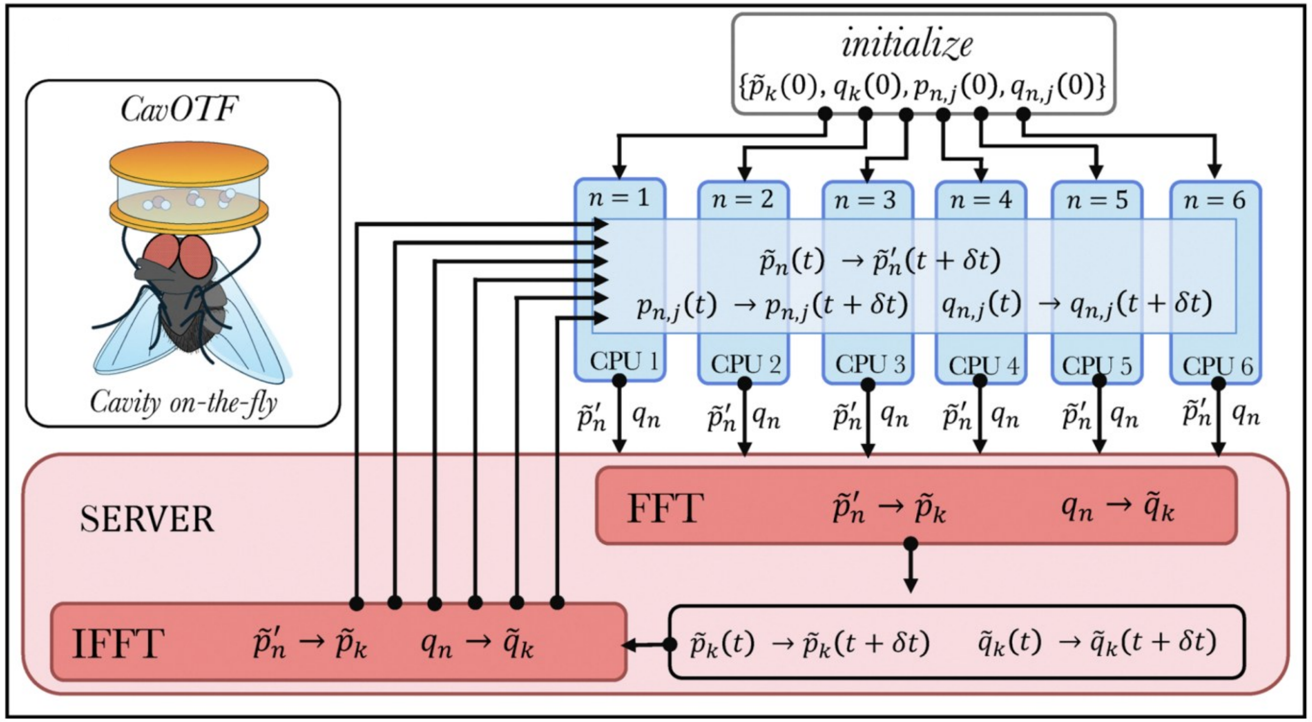
$$\approx \widehat{H}_{\mathrm{M}} + \sum_{k} \omega_k \widehat{a}_k^\dagger \widehat{a}_k + \sqrt{2N} g_0 \omega_0 \sum_{n} \widehat{q}_n \widehat{M}_n + N g_0^2 \sum_{n} \widehat{M}_n^2$$

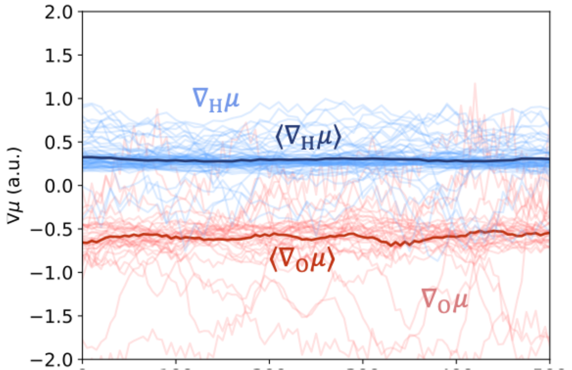


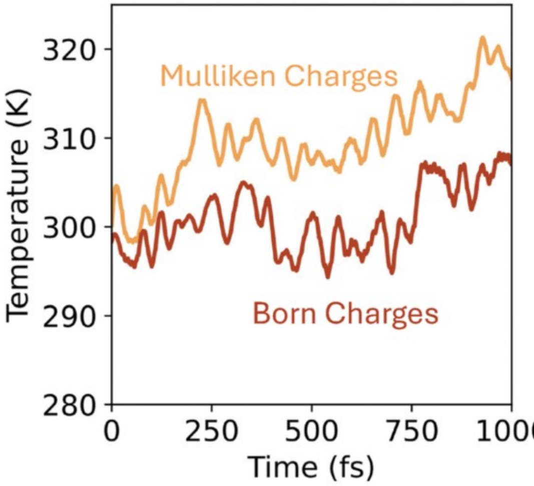
$$\hat{M}_n = (\hat{\varepsilon} \cdot \hat{\mu}_n)$$

cavOTF: on-the-fly cavity-molecular dynamics

$$\hat{H}_{\mathrm{LM}} = \hat{H}_{\mathrm{M}} + \sum_k \omega_k \hat{a}_k^\dagger \hat{a}_k + \eta \sum_n \hat{q}_n \hat{\mu}_n + \frac{\eta^2}{2\omega_0{}^2} \sum_n \hat{\mu}_n{}^2$$





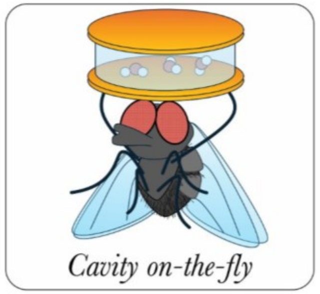


cavOTF

On-the-fly cavity–molecular dynamics for vibrational polaritons

Sachith Wickramasinghe, Amirhosein Amini, Arkajit Mandal, PCCP (2026)

pypi v0.1.7.12.25 | Python ≥3.9 | license not specified | DOI 10.1039/D5CP04879F

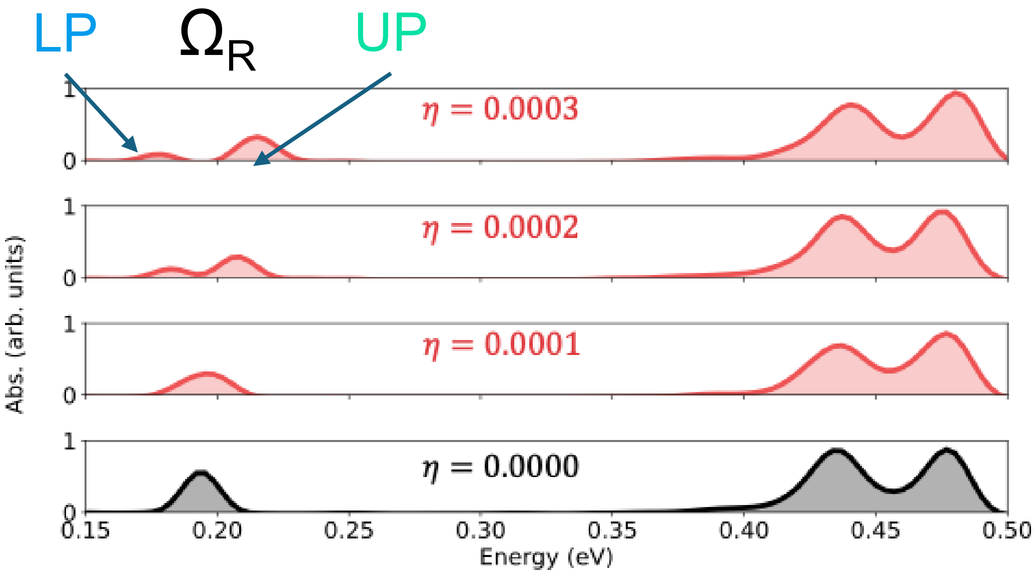


DFTB with SCC

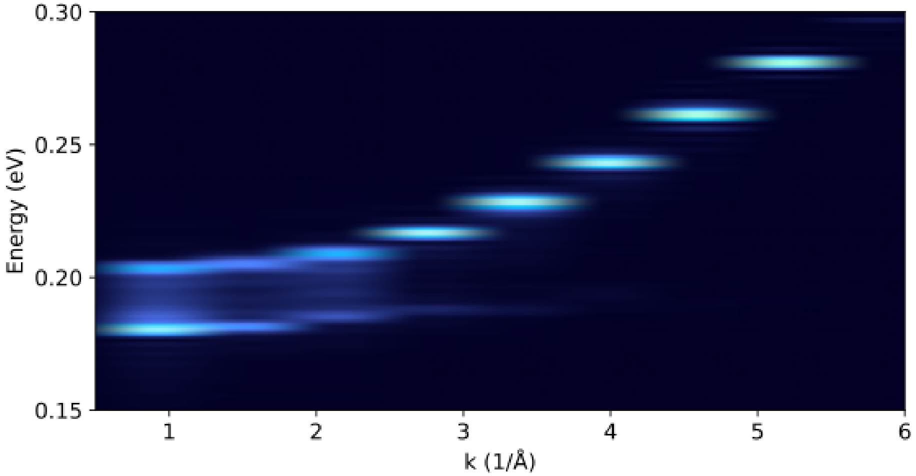
Wickramasinghe, Amini, Mandal **Phys. Chem. Chem. Phys.** (2026)

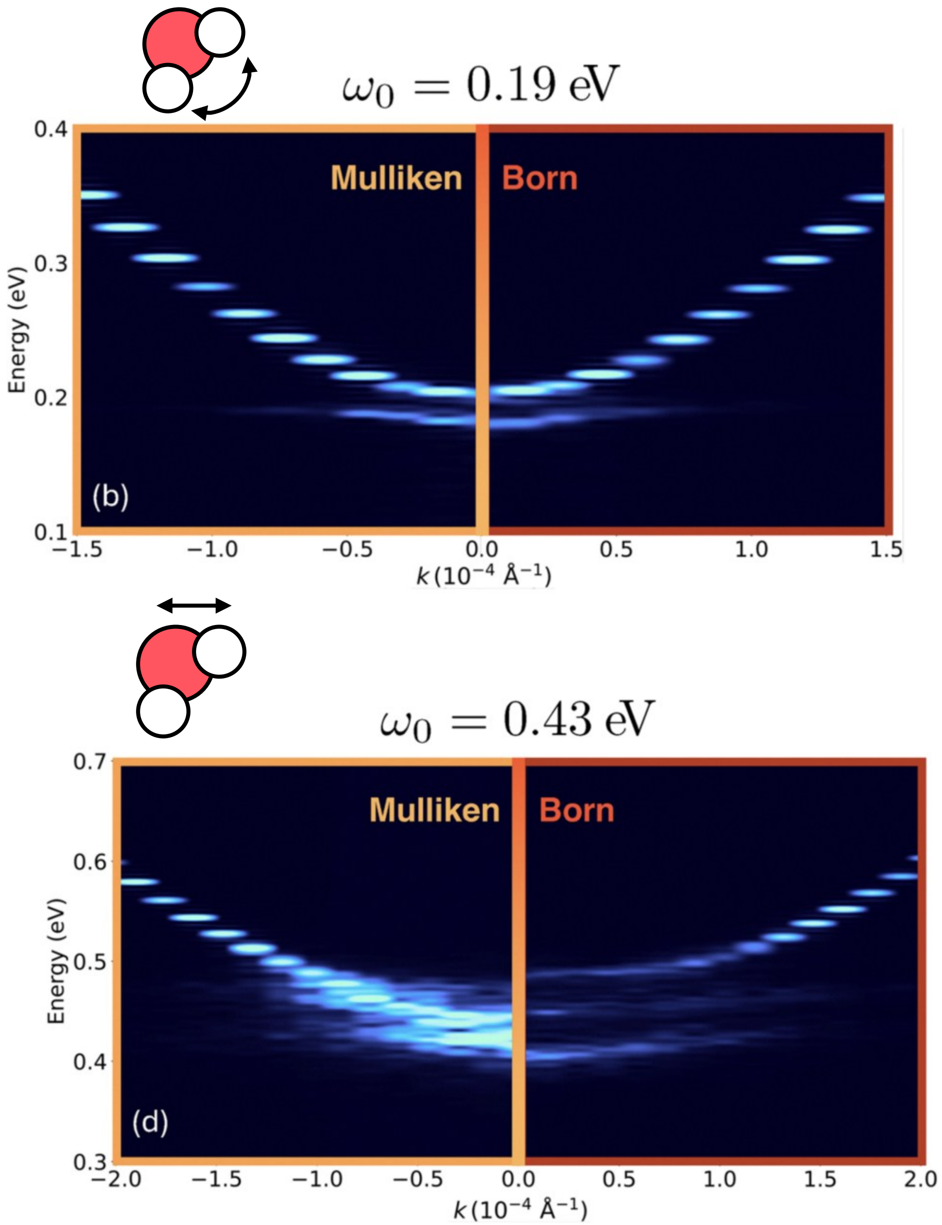
cavOTF: Captures Vibrational-Polariton Formation in Water

Simulated IR spectra



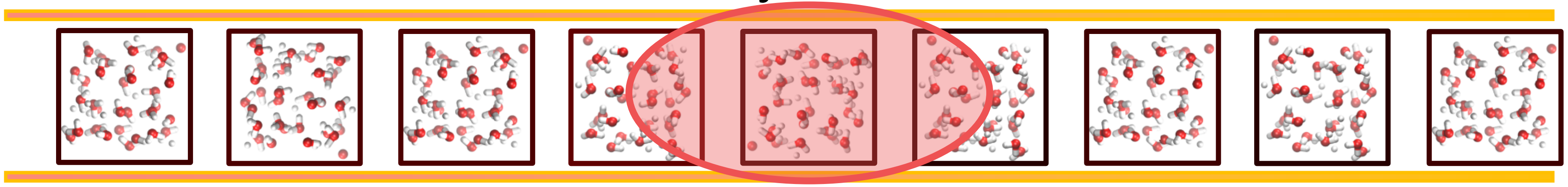
Simulated angle resolved spectra



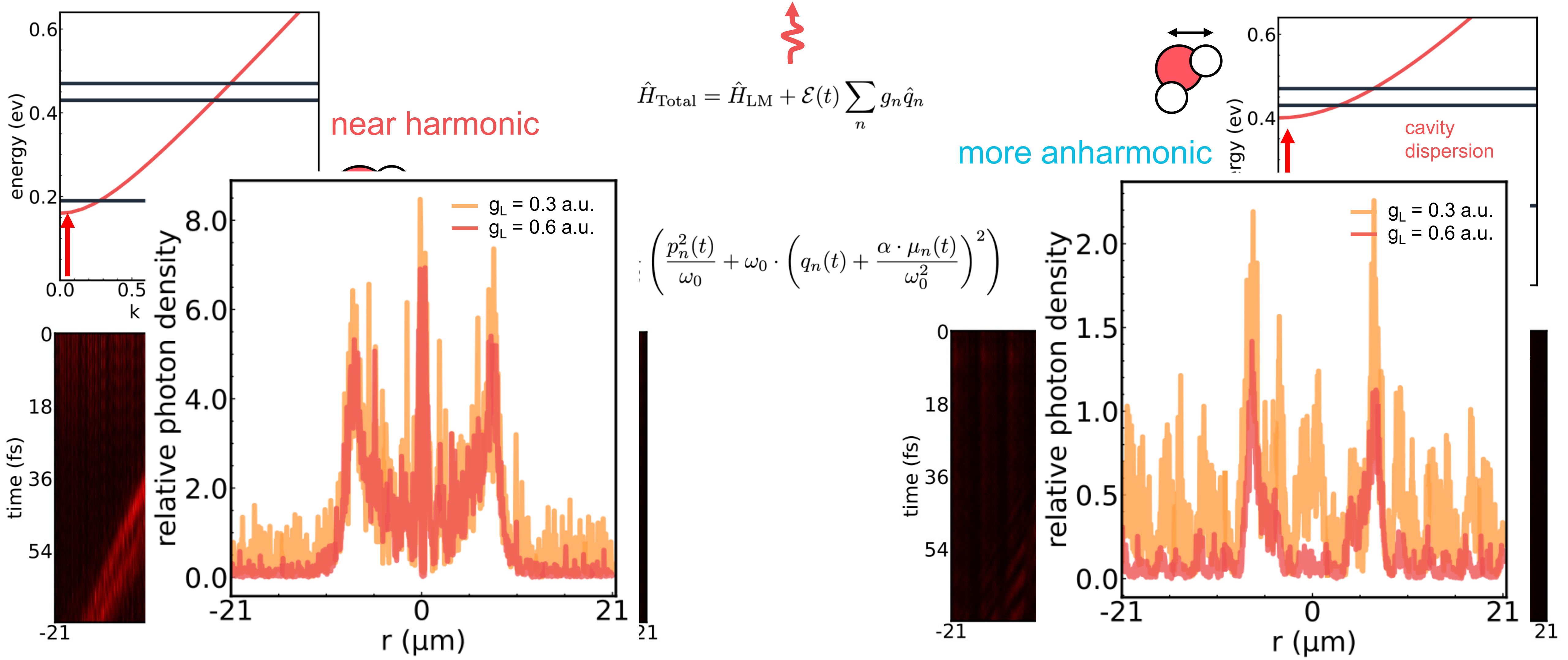


Wickramasinghe, Amini, Mandal **Phys. Chem. Chem. Phys.** (2026)

Molecular Anharmonicity Controls Photonic Localization

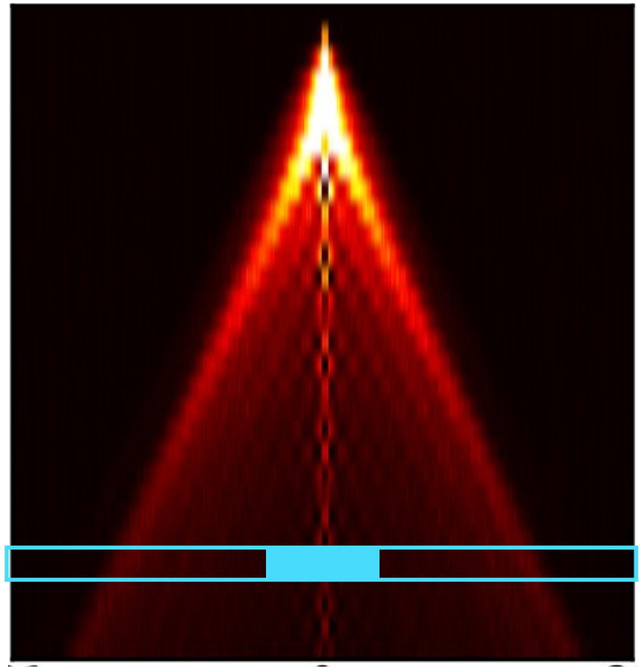


$$\hat{H}_{\text{Total}} = \hat{H}_{\text{LM}} + \mathcal{E}(t)\sum_{n} g_n \hat{q}_n$$

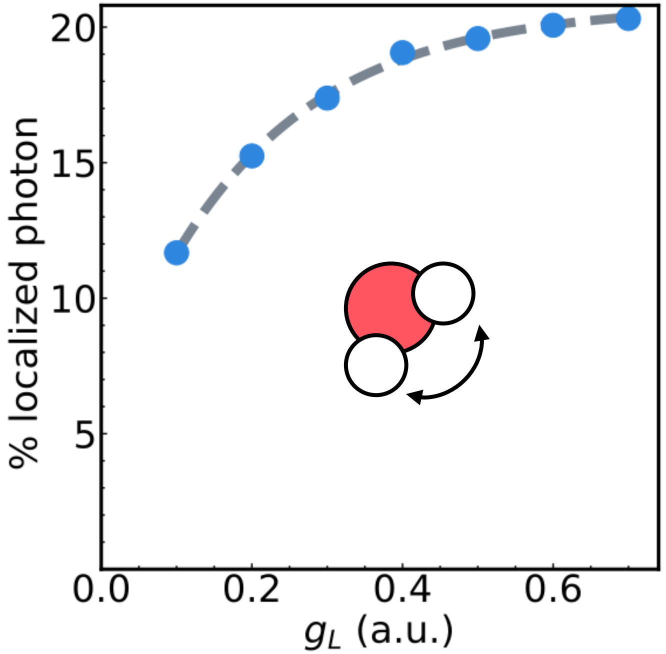


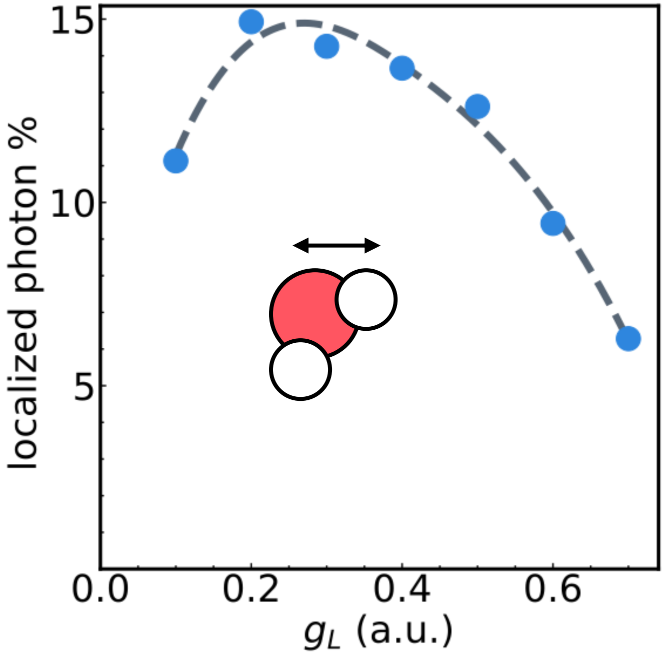
$$\left(\frac{p_n^2(t)}{\omega_0} + \omega_0 \cdot \left(q_n(t) + \frac{\alpha \cdot \mu_n(t)}{\omega_0^2}\right)^2\right)$$

Molecular Anharmonicity Controls Photonic Localization



$$\mathcal{N}_{loc} = \frac{\int_{t-\Delta t}^{t+\Delta t} d\tau \int_{-r_c}^{+r_c} dr N(r,\tau)}{\int_{t-\Delta t}^{t+\Delta t} d\tau \int_{-\infty}^{+\infty} dr N(r,\tau)}$$



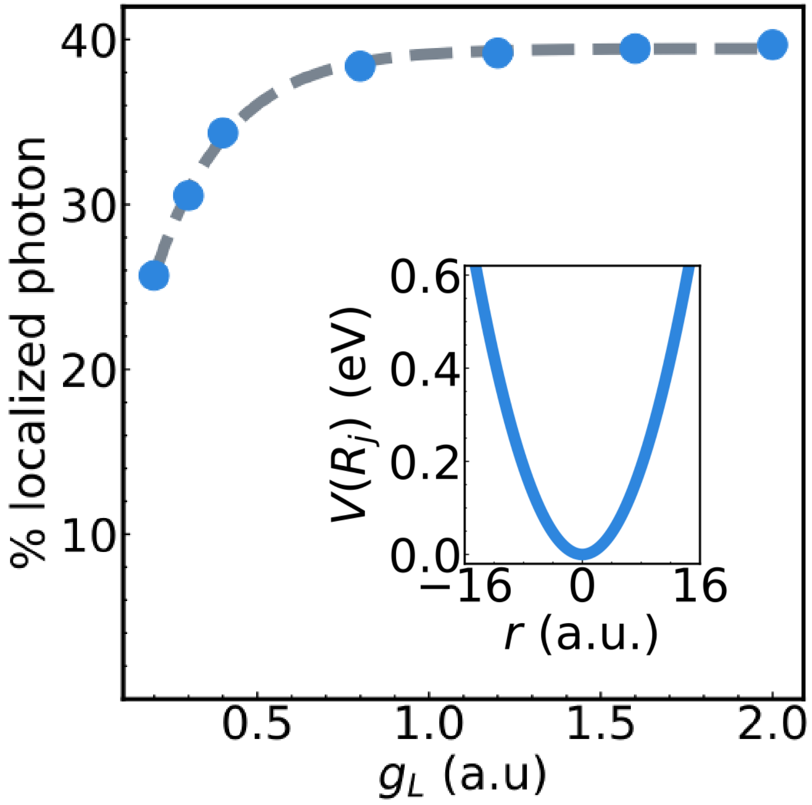


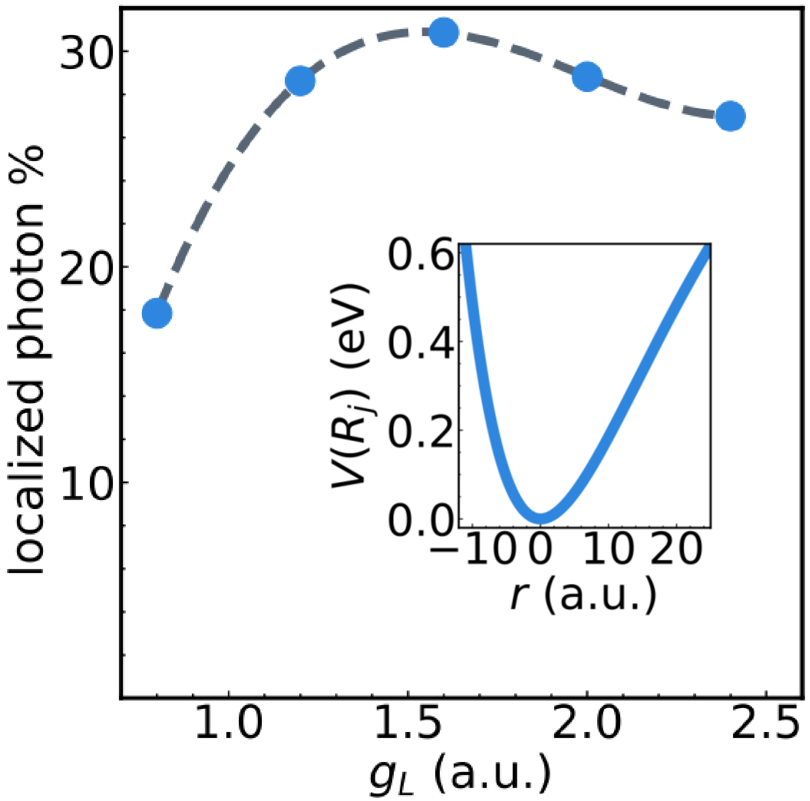
As g_L is increased, the localized photonic density overcomes the thermal noise, and we consequently observe an increase in the % photonic localization

near harmonic mode, localization increases and saturates
near anharmonic mode localization becomes non-monotonic

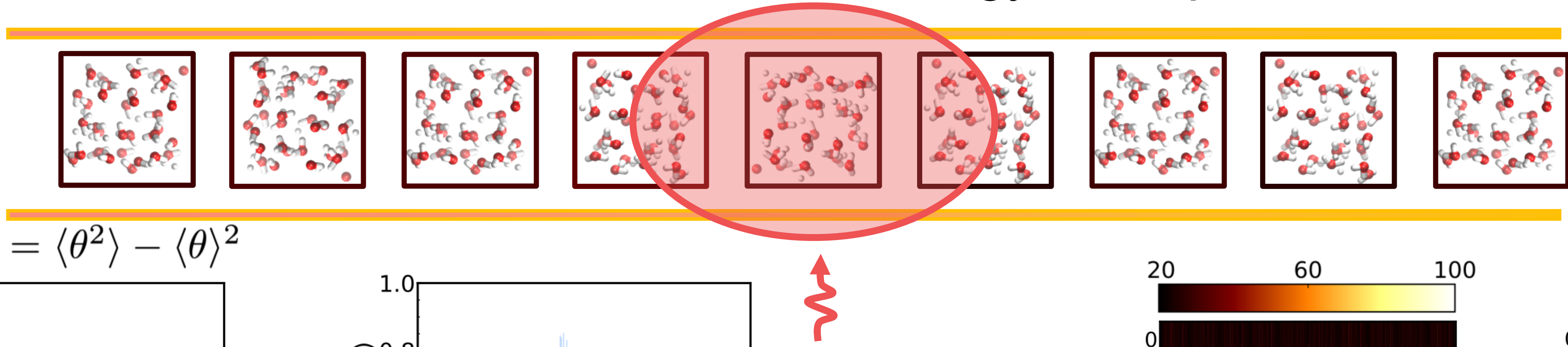
Molecular Anharmonicity Controls Photonic Localization

$$H_{\mathrm{LM}} = H_{\mathrm{M}} + \frac{1}{2}\sum_k \left(p_k^2 + \omega_k^2 q_k^2\right) + \sum_n q_n \mu_n + \frac{1}{2\omega_0^2}\sum_n \mu_n^2$$

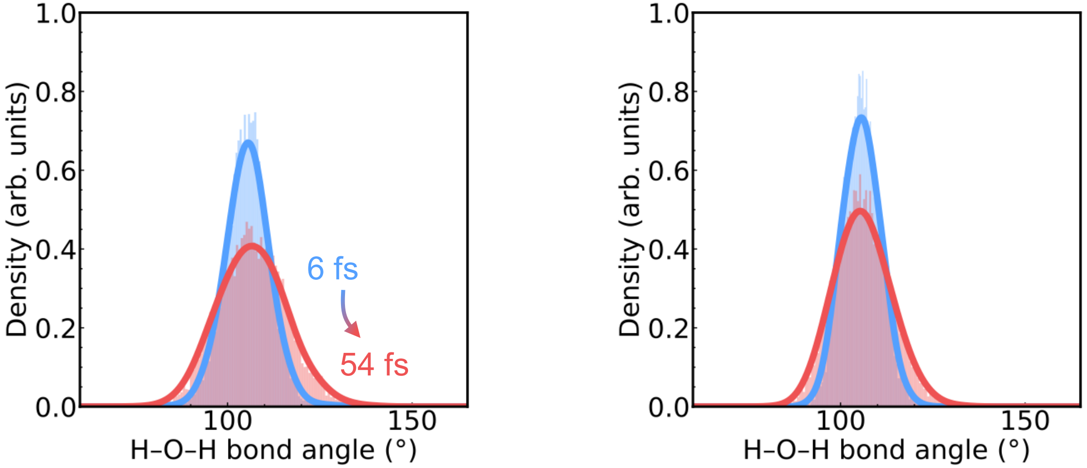




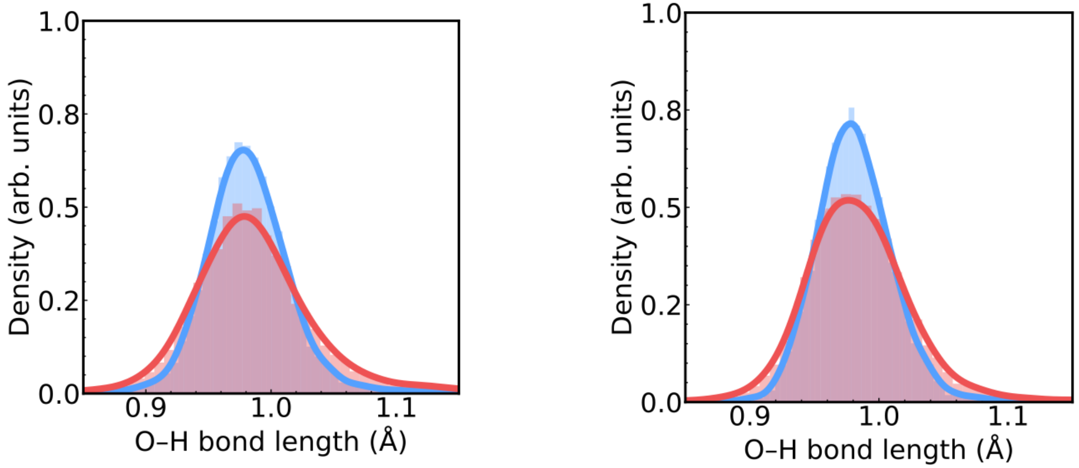
Mode-Selective Energy Transport

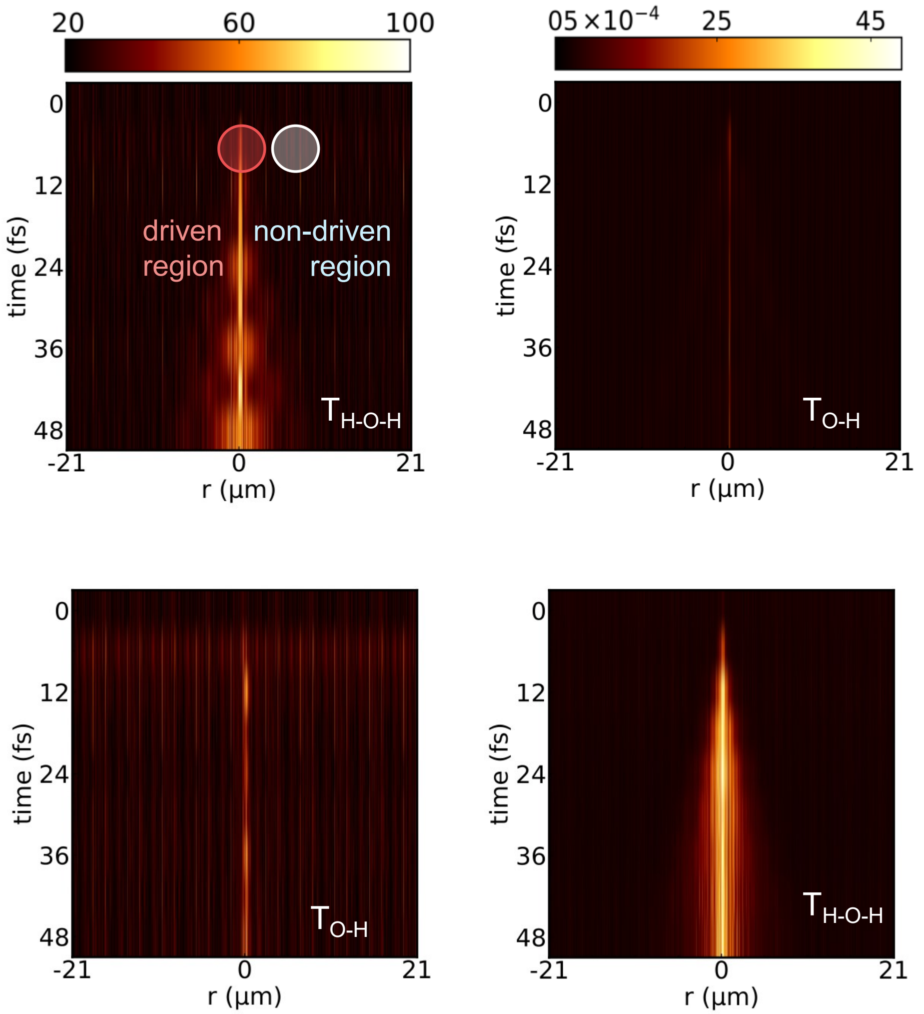


$$T_{\mathrm{H-O-H}} = \langle \theta^2 \rangle - \langle \theta \rangle^2$$

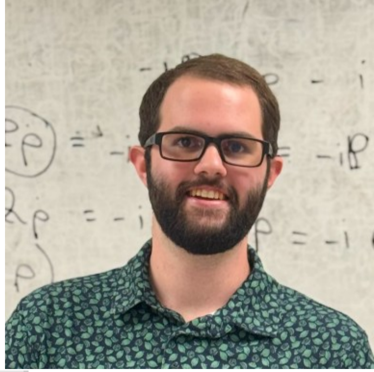


$$T_{\mathrm{O-H}} = \langle r^2_{\mathrm{O-H}} \rangle - \langle r_{\mathrm{O-H}} \rangle^2$$

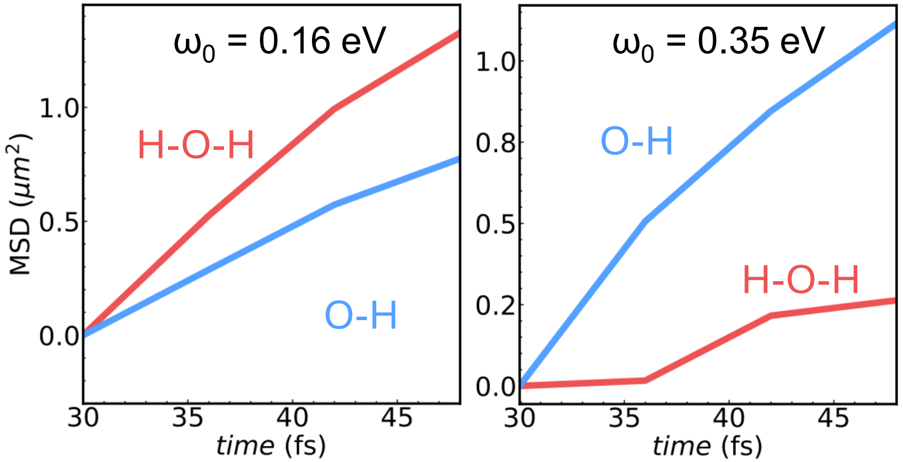


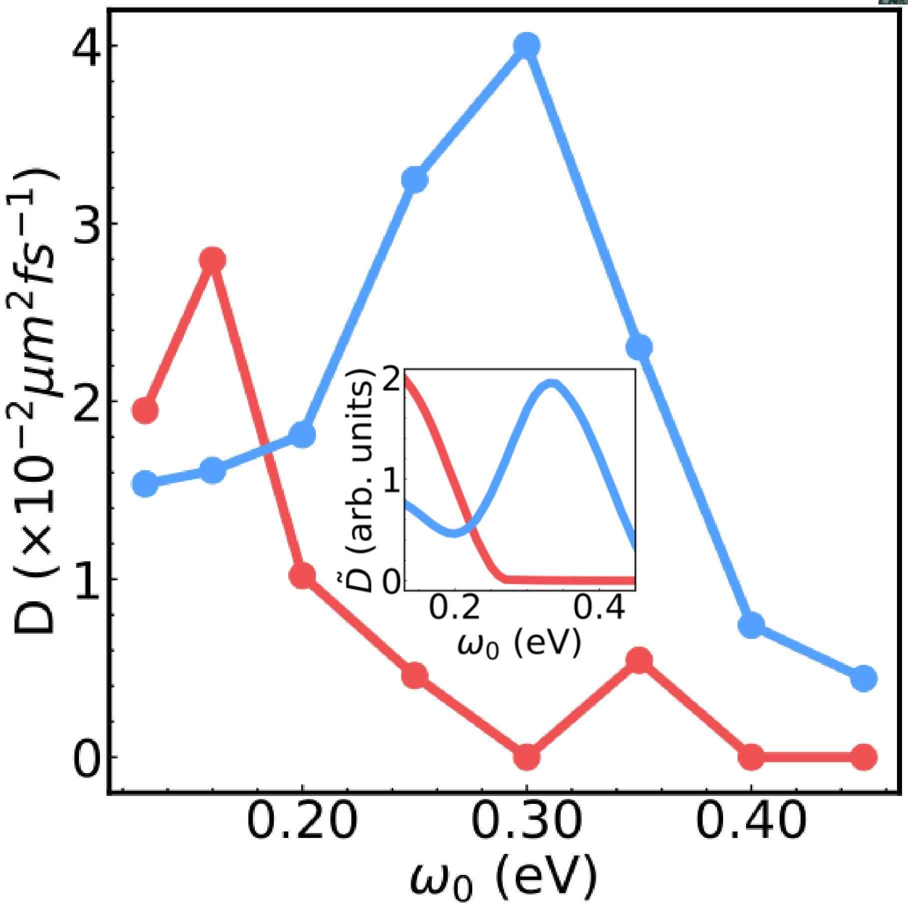


Mode-Selective Energy Transport



$$\mathrm{MSD}(t) = \frac{\int_{-\infty}^{\infty} dr\ r^2 T_j(r, t+t_s)}{\int_{-\infty}^{\infty} dr T_j(r, t+t_s)} - \frac{\int_{-\infty}^{\infty} dr\ r^2 T_j(r, t_s)}{\int_{-\infty}^{\infty} dr T_j(r, t_s)}$$





$$\tilde{D}(\omega) = \frac{I}{\Delta\omega} \int_{\omega-\Delta\omega/2}^{\omega+\Delta\omega/2} d\omega v_k^2(\omega) \cdot |\mathcal{E}(\omega)|^2$$

Wickramasinghe, Fowler, Groenhof, Mandal **arXiv** (2026)

Takeaways

Mesoscale simulations beyond the long-wavelength approximation are essential for capturing spatially resolved vibro-polariton dynamics

Mulliken charges alone are insufficient to describe the complex light–matter dynamics

Molecular anharmonicity leads to non-monotonic photon localization, particularly as the driving strength is increased

Cavity frequency provides a control knob for energy transport and photon localization by tuning the interplay between molecular vibrations and photonic modes

Acknowledgement

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ACCESS

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